SCI-FLT-FIXED v0.1 Engineering-Conformance Specification

SCI-FLT-FIXED-ENGINEERING-CONFORMANCE v0.1/draft-r0.1

Status: implementation-blind Stage B conformance-target draft; no conformity finding

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Document identity: SCI-FLT-FIXED-ENGINEERING-CONFORMANCE v0.1/draft-r0.1
Source SHA-256: 65c90bdc1d3c1ea486b364c3c5a7fbc5b1f2d0b3b05e33a9b034b285209908df
Shared normative core SHA-256: 867c48304906c43da1ac6f54893fe5988e7fbbeecde393519cdd7b0ea4f7a97f
Author packet manifest SHA-256: 7f2d03f182258ac9770f7dba869e9ae0b5018efdcdb93b18b299a9b9c6df1e4d
Build recipe SHA-256: ae633e16250ce4e3d6aa95e53555327676aa75f4213f32104391e799207217c8
```

Implementation-blind scientific-contract draft. This artifact reports no implementation, validation, calibration, achieved-response, achieved-covariance, performance, readiness, freeze, production, or Unity result.

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Normative import: the complete `SCI-FLT-FIXED-NORMATIVE-CORE v0.1/draft-r0.1`, source SHA-256 `867c48304906c43da1ac6f54893fe5988e7fbbeecde393519cdd7b0ea4f7a97f`, is incorporated without modification. This view adds no scientific rule. If this view and the imported core differ, the core controls.

1. Conformance target and evidence boundary

This specification translates the shared scientific core into observable engineering obligations without describing, inspecting, or judging an implementation. A future conformance review must bind one candidate realization to one exact core revision, parent generation, resolved operator generation, and product generation.

A claim passes only when positive evidence establishes the exact required identity and behavior. A missing, conflicting, approximate, same-name, or inferred fact yields unavailable or failed status as directed by the core. A finite array, successful process exit, or familiar label is not sufficient.

Conformance, scientific validation, calibration, performance, readiness, and production are separate evidence classes. This specification defines only a future conformance target.

2. Required input and plan records

A candidate realization must accept an immutable request record, effective decision, complete parent binding, and externally resolved fixed plan. The records must expose enough content to reconstruct and compare exact identities, not merely display friendly names.

The parent record must distinguish MAP observation, MAP coadd, and JINC observation roles. It must bind the complete applicable upstream identity, quantity, units, nominal-beam, WCS, support, validity, response, covariance, null, exposure, lifecycle, failure, and provenance state. JINC requires all five atomic numerical roles. Upstream unavailability must remain explicit.

The plan record must distinguish requested, effective, disabled, unavailable, and resolved states. A resolved plan must bind every coefficient, parameter, grid and metric fact, normalization, support and edge rule, transfer qualification, and provenance fact before any parent or NOI member is applied.

3. Operator reconstruction record

`FLT-OPERATOR` must provide a content-bound representation sufficient to reconstruct the exact finite `J_full L_Theta`, including:

- ordered input and output row domains;
- exact WCS, frame, topology, metric, shape, indexing, and pixel-area facts;
- complete sampled coefficients, coefficient units and ordering, numerical representation, and digest;
- kernel offset set, center, extent, tie, phase, subpixel convention, orientation, and handedness;
- declared normalization, DC gain, and distinct signed, absolute, squared, and geometric support summaries;
- full-footprint predicates and cause vocabulary;
- qualified transfer facts or explicit unavailability; and
- immutable plan and operator generation identifiers.

An alternate computational mechanism is comparable only after reconstructing the identical declared operator. A different coefficient representation that changes a declared coefficient, ordering, phase, row domain, or numerical operator is not scientific equivalence.

4. Admission and application gate

Input admission must evaluate the draft `SCI-FLT-FIXED:input_admission@1` semantics against one exact request, parent, and resolved plan. It must return distinct request, applicability, eligibility, unavailable-decision, and cause state. It performs no transformation.

Application may proceed only for an eligible resolved route. For every output row, the row selector must verify all required kernel locations against the exact parent domain, FLT input admission, payload availability, finiteness, and all required predicates. The scientific row domain must equal the set of rows that pass every check.

No extension, periodic wrap, truncation, local renormalization, inpainting, reflection, clamp, mirror, padding-based admission, edge completion, or value replacement is conforming in v0.1. Rows failing the full-footprint gate remain unavailable with causes, even if a storage payload is finite.

5. Signal, response, transfer, mode, and influence records

Application must compute `FLT-SIG` from the exact resolved operator on the exact scientific row ordering. The output-unit record must derive units from parent and coefficient units and retain the originating nominal-beam and calibration lineage without introducing a filtered-beam label.

When an exact compatible parent response is available, the response record must equal the identical operator applied to that response on the identical row domain. Otherwise the record must be explicitly unavailable. Kernel-only, approximate, differently centered, differently normalized, or differently edged response surrogates do not pass.

Transfer and mode records must bind the exact finite-grid domain, phase, normalization, null, invariant, and attenuation facts that are scientifically defined. Unsupported facts must remain unavailable. Influence must expose the parent-row coefficient relation and must remain separate from physical exposure.

6. Covariance and NOI records

For an available compatible parent covariance, a covariance-qualified candidate must realize the exact two-sided operator propagation on `S_out`. Its record must state whether it is complete, diagonal-input propagated, structured, partial, or marginal; name all omitted terms; and expose domain, ordering, rank, null space, and supported operations where applicable.

If an explicitly diagonal parent covariance is propagated, any complete output representation must include induced cross-row covariance. A marginal-only representation must say so. Missing cross terms may not be encoded as zero or independence.

An optional NOI attachment must be separately owned and immutably bound to the exact FLT product. Every admitted NOI member must receive the identical operator state and row selection as the signal. The route must reject filtering of uncertainty-like products, approximate transfer, relocation, commutation, substitution, and any per-member re-resolution.

7. Atomic publication and lifecycle records

Publication must evaluate the draft `SCI-FLT-FIXED:output_publication@1` semantics against one complete bundle. The bundle must contain all required roles from the imported core, including honest unavailable companion records where permitted. A partial bundle, placeholder, or inferred companion fails required publication.

Lifecycle evidence must distinguish `not_requested`, `requested`, `effective`, `disabled`, `unavailable`, `resolved`, `applied`, `failed`, `realized_identity`, `realized_zero`, `realized`, and `superseded`. Disabled emits no product. Identity and zero are realized transformations. Required failure propagates and emits no complete product.

Every product must bind immutable parent, plan, operator, output, companion, cause, lifecycle, failure, and provenance generations. Any core-defined identity change creates a new generation. A later NOI attachment is a separate companion and cannot mutate an existing FLT bundle.

8. Requirement conformance matrix

The following matrix routes every stable core requirement to a future observable conformance decision. The exact normative text remains in the imported core.

- `SCI-FLT-FIXED-REQ-001`: verify package and tranche identity; reject generic or inference-bearing identity.
- `SCI-FLT-FIXED-REQ-002`: reconstruct A and verify no additive term or additive-state dependency.
- `SCI-FLT-FIXED-REQ-003`: verify exactly one supported immutable parent role.
- `SCI-FLT-FIXED-REQ-004`: verify complete applicable parent fields and atomic JINC roles.
- `SCI-FLT-FIXED-REQ-005`: verify upstream unavailability remains fail-closed.
- `SCI-FLT-FIXED-REQ-006`: verify observation, coadd, and JINC successor identities and absence of FLT coaddition or inferred commutation.
- `SCI-FLT-FIXED-REQ-007`: verify all operator state predates application and is independent of parent and member payloads.
- `SCI-FLT-FIXED-REQ-008`: compare exact WCS and grid facts and reject approximate joins or resampling.
- `SCI-FLT-FIXED-REQ-009`: reconstruct exact finite coefficients, domains, discretization, and digest.
- `SCI-FLT-FIXED-REQ-010`: verify convolution construction from the exact sampled kernel and offset set.
- `SCI-FLT-FIXED-REQ-011`: verify every low-pass transfer fact or mark only the qualification unavailable.
- `SCI-FLT-FIXED-REQ-012`: recompute the full-footprint row set from exact predicates.
- `SCI-FLT-FIXED-REQ-013`: verify all excluded rows are unavailable with causes, never promoted by storage shape.
- `SCI-FLT-FIXED-REQ-014`: detect and reject every deferred edge or replacement method.
- `SCI-FLT-FIXED-REQ-015`: verify transformed-amplitude identity and exact unit derivation without amplitude-category relabeling.
- `SCI-FLT-FIXED-REQ-016`: verify nominal-beam and CAL lineage retention and absence of unsupported beam or flux claims.
- `SCI-FLT-FIXED-REQ-017`: compare response output to exact operator composition or verify honest unavailability.
- `SCI-FLT-FIXED-REQ-018`: verify finite-domain transfer, mode, and phase facts or explicit unavailable states without whole-chain promotion.
- `SCI-FLT-FIXED-REQ-019`: compare influence to coefficients and verify it is not labeled exposure.
- `SCI-FLT-FIXED-REQ-020`: verify distinct computability, support, admission, validity, confidence, and downstream states.
- `SCI-FLT-FIXED-REQ-021`: compare covariance to exact two-sided propagation.
- `SCI-FLT-FIXED-REQ-022`: verify covariance representation type, omissions, domain, and unknown cross-term handling.
- `SCI-FLT-FIXED-REQ-023`: test induced off-diagonal terms and reject marginal planes presented as complete covariance.
- `SCI-FLT-FIXED-REQ-024`: verify empirical uncertainty and significance are absent from FLT ownership.
- `SCI-FLT-FIXED-REQ-025`: compare every NOI member's full operator state to the signal state.

- `SCI-FLT-FIXED-REQ-026`: detect and reject per-member selection or re-resolution.
- `SCI-FLT-FIXED-REQ-027`: verify every atomic role and allowed honest unavailable record.
- `SCI-FLT-FIXED-REQ-028`: verify the complete lifecycle vocabulary, causes, failure, and generation bindings.
- `SCI-FLT-FIXED-REQ-029`: test distinct disabled, identity, and zero outcomes.
- `SCI-FLT-FIXED-REQ-030`: mutate each identity-defining fact in isolation and verify a new immutable generation; verify NOI nonmutation.
- `SCI-FLT-FIXED-REQ-031`: remove or conflict each required fact and verify fail-closed behavior without fallback.
- `SCI-FLT-FIXED-REQ-032`: evaluate all input-admission request, applicability, eligibility, unavailable, exclusion, and cause branches.
- `SCI-FLT-FIXED-REQ-033`: evaluate complete, honest-unavailable, partial, disabled, identity, zero, and failed publication branches.
- `SCI-FLT-FIXED-REQ-034`: verify VAL only binds or evaluates an immutable approved policy and authors no FLT fact or arithmetic.
- `SCI-FLT-FIXED-REQ-035`: verify no generic downstream admission follows from product availability.
- `SCI-FLT-FIXED-REQ-036`: verify every excluded family and every Stage B nonclaim remains outside the product claim set.

9. Falsifiable prediction suite

Each future prediction evaluation must bind exact inputs, operator state, expected row domain, expected companions, expected lifecycle, comparison semantics, and causes before execution. The normative expected outcomes are in the imported core.

- `SCI-FLT-FIXED-PRED-001`: identity operator and identity lifecycle.
- `SCI-FLT-FIXED-PRED-002`: zero operator, zero signal, honest unavailable companions, and zero lifecycle.
- `SCI-FLT-FIXED-PRED-003`: scalar input linearity.
- `SCI-FLT-FIXED-PRED-004`: constant input under exact DC gain and every authorized normalization.
- `SCI-FLT-FIXED-PRED-005`: impulse, center, orientation, phase, and indexing.
- `SCI-FLT-FIXED-PRED-006`: exact compatible parent-response composition.
- `SCI-FLT-FIXED-PRED-007`: signed coefficients and distinct support summaries.
- `SCI-FLT-FIXED-PRED-008`: zero-sum constant-mode null without claim promotion.
- `SCI-FLT-FIXED-PRED-009`: full-footprint inclusion and dependent-row removal.
- `SCI-FLT-FIXED-PRED-010`: rejection of every deferred edge and replacement method, including support-conditioned renormalization.
- `SCI-FLT-FIXED-PRED-011`: missing, unavailable, non-admitted, and non-finite required parent locations.
- `SCI-FLT-FIXED-PRED-012`: exact complete covariance on ordered `S_out`.
- `SCI-FLT-FIXED-PRED-013`: induced cross-row covariance from diagonal input.
- `SCI-FLT-FIXED-PRED-014`: unavailable parent response or covariance and rejection of surrogates.
- `SCI-FLT-FIXED-PRED-015`: exact WCS and grid mismatch failure.
- `SCI-FLT-FIXED-PRED-016`: distinct observation, coadd, and JINC identities with no FLT coadd product.
- `SCI-FLT-FIXED-PRED-017`: exact fixed-state NOI member parity.

- `SCI-FLT-FIXED-PRED-018`: per-member re-resolution rejection.
- `SCI-FLT-FIXED-PRED-019`: disabled, identity, zero, and failure lifecycle distinctions.
- `SCI-FLT-FIXED-PRED-020`: upstream unavailable MAP and JINC route.
- `SCI-FLT-FIXED-PRED-021`: low-pass qualification completeness.

10. Traceability, identity, and reproducibility record

`TRACEABILITY.json` maps every requirement and prediction identifier to its normative-core section, scientist-rationale section, engineering-conformance section, and exact admitted Stage A object sections. The durable verifier must reject missing, duplicate, extra, malformed, or unadmitted trace entries.

`BUILD_BINDING.json` records the packet manifest binding, all 17 admitted object hashes, every Stage B source and build-tool hash, deterministic build environment, output PDF identities, page counts, byte sizes, and SHA-256 digests. A clean rebuild in a separate temporary directory must reproduce the three PDF digests exactly.

PDF text must expose the document identity, exact source digest, exact shared core digest, exact packet-manifest digest, and exact builder digest. PDF page rendering and visual QA are required in addition to content and hash checks.

11. Nonclaims

This conformance specification does not identify, inspect, modify, or judge an implementation. It records no implementation conformity, algorithm change, validation result, calibration state, achieved response or covariance, numerical adequacy, performance, readiness, scientific freeze, production status, or Unity activity.